

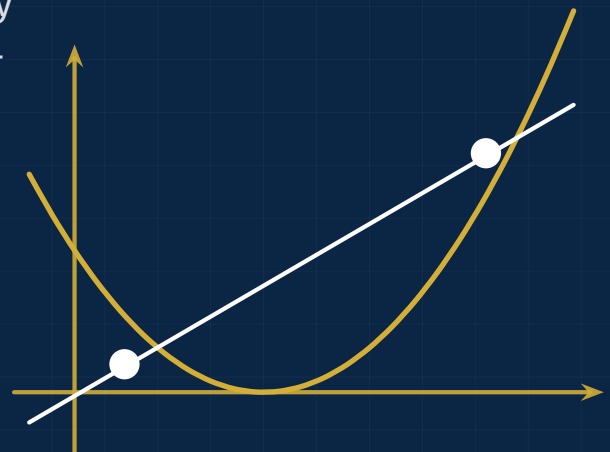


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 6

Estimating Gradients

A curve has a different gradient at every point. Draw the tangent, make a big triangle, rise over run — then say what it means.



4

VARIANTS

4

PAST-PAPER PARTS

22

PRACTICE QUESTIONS

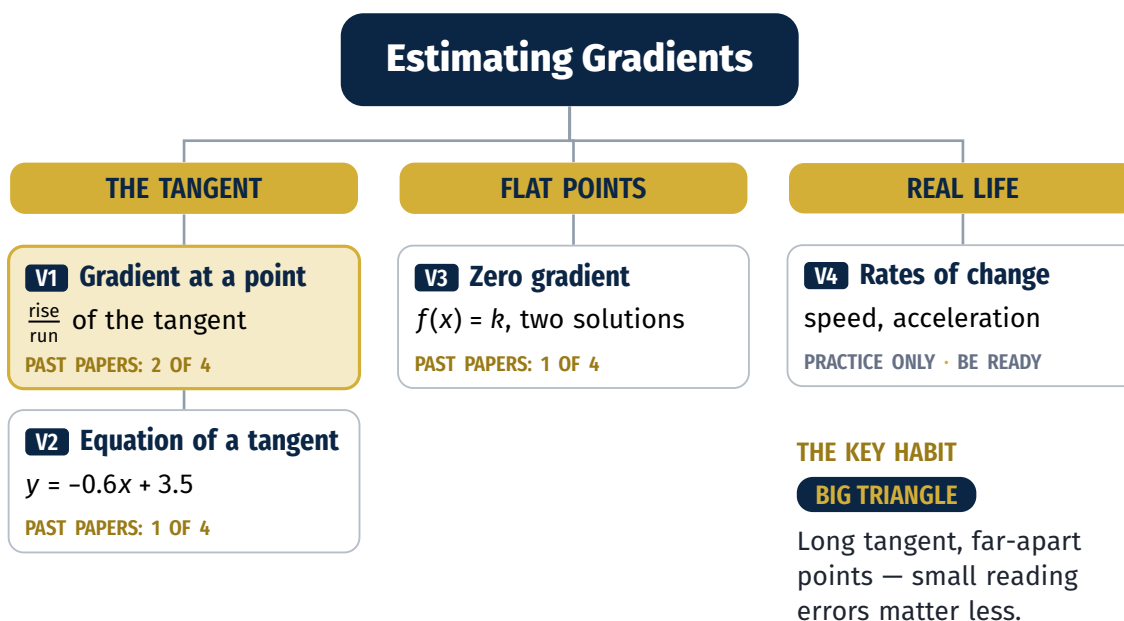
YOU WILL MASTER

V1 Gradient at a point

V2 Equation of a tangent

V3 Zero gradient

V4 Rates of change



★ How often does each variant come up?

		PAST PAPERS (OUT OF 4)		PRACTICE PARTS	
V1	Gradient at a point		2		4
V2	Equation of a tangent		1		1
V3	Zero gradient		1		2
V4	Rates of change		0		15

WHAT THIS TELLS YOU Rare but valuable (3–7 marks). The answer is an estimate, so the marks are for the tangent and the working: draw it, show the triangle, write $\text{rise} \div \text{run}$.

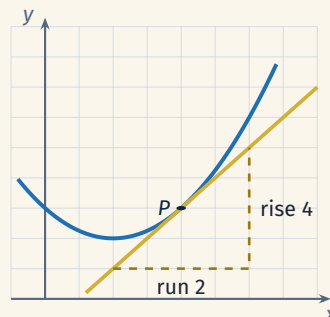
BEFORE YOU START

The big idea

The gradient of a curve at a point = the gradient of its tangent there.

A **tangent** is a straight line that just touches the curve at that point, with the same steepness.

Draw it with a ruler, then measure its gradient like any line.



 Your notes

What the gradient means

Distance–time

$$\frac{\text{m}}{\text{s}}$$

Gradient = speed.

Velocity–time

$$\frac{\text{m/s}}{\text{s}}$$

Gradient = acceleration.

Zero

flat

Top or bottom of a curve.

Negative



The quantity is falling.

Tangent or chord?

Tangent at one point

“at $t = 5$ ”

Instantaneous rate — the gradient at that moment.

Chord between two points

$$\frac{30-3}{40-0}$$

Average rate over the interval.



CALCULATOR FIRST, THEN CHECK

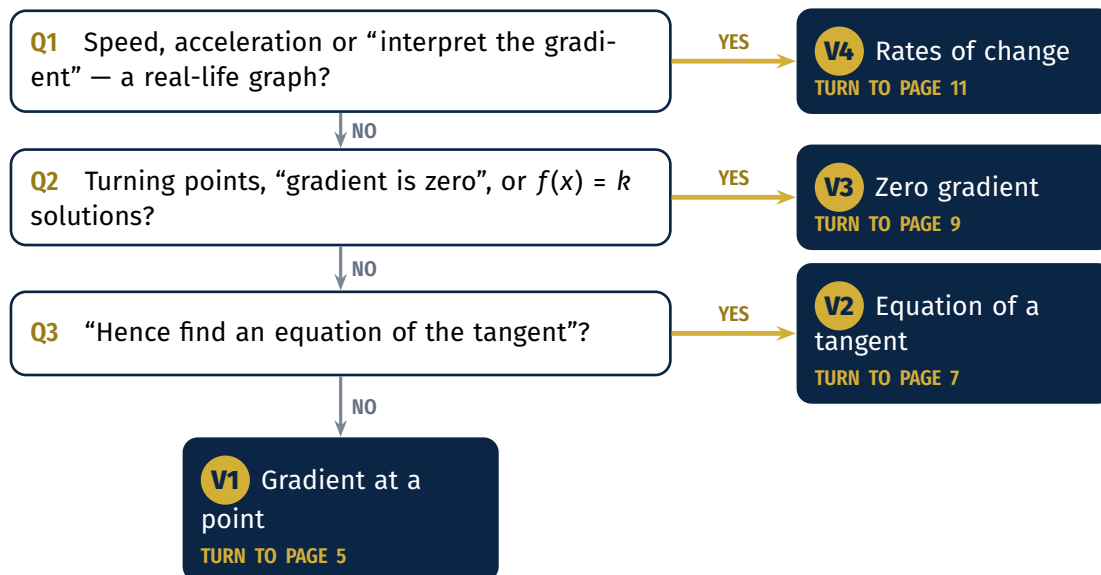
Units of the gradient = (y units) per (x units). Always put them in a real-life answer.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1 MOST COMMON IN THE EXAM

Gradient at a Point

Rule the tangent at the point, pick two far-apart points on it, rise over run.

SEEN IN PAST PAPERS

● ● ● ● 2 / 4

PRACTICE QUESTIONS: 4



Your notes

Came up in: Jan 2020 P1HR Q18(a)(i) · Jan 2022 P2HR Q17

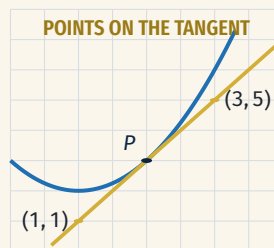
? What is it?

In plain words

Place the ruler so it **just touches** the curve at the point — same steepness on both sides. Draw it **long**.

Choose two points **on the tangent** (not the curve), far apart, on grid lines. Gradient = $\frac{\text{rise}}{\text{run}}$; downhill is **negative**.

IT LOOKS LIKE

at $x = 2$ at P 

How do I do it?

1 Tangent

Ruler touching at the point only; draw it long.

2 Two points

On the tangent, far apart, on grid lines.

3 Rise, run

Use the axis scales.

4 Divide

 $\frac{\text{rise}}{\text{run}}$; check the sign.**BIG TRIANGLE**

It is an estimate — show the triangle.

V1 Gradient at a Point

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find an estimate for the gradient of the curve at the point where $x = 2$ ”

→ **Tangent at $x = 2$; rise over run**

“Show your working clearly”

→ **Draw the tangent and the triangle on the grid**

Curve going down at the point

→ **Negative answer****WORKED EXAMPLE****PRACTICE**

Practice · Medium · Q5(b) · 3 marks

P is the point on $y = x^2 - 2x + 3$ where $x = 2$. Calculate an estimate for the gradient of the graph at P .

STEP 1 $P(2, 3)$

◀ the point

STEP 2tangent through $(1, 1)$ and $(3, 5)$

◀ rule the tangent; read two points

STEP 3

rise = 4, run = 2

◀ triangle

STEP 4 $m = \frac{4}{2} = 2$

◀ divide

ANSWER about 2✓ curve rising at P : positive ✓**WATCH OUT**

✗ using two points on the curve

✓ two points on the tangent

Points on the curve give a chord — an average, not the gradient at P .

**EXAM TIP**

A small range of answers is accepted, but only if the tangent really touches.

**Check yourself**

calculator first · answers on page 13

1.1

Jan 2022 P2HR

The tangent at $x = 2$ passes through $(1.5, -0.25)$ and $(3, 3.8)$. Estimate the gradient.

Answer:

1.2

Jan 2020 P1HR

The tangent at $P(2, 2.3)$ passes through $(6, -0.1)$. Estimate the gradient at P .

Answer:

1.3

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The tangent to a curve at $P(3, 5)$ also passes through $(0, -1)$. Find the gradient at P .

Answer:

V2

VARIANT 2

Equation of a Tangent

The tangent is a straight line: its gradient (V1) and the point P give $y = mx + c$.

SEEN IN PAST PAPERS

● ● ● ● 1 / 4

PRACTICE QUESTIONS: 1



Your notes

Came up in: Jan 2020 P1HR Q18(a)(ii)

? What is it?

In plain words

You already know the tangent's **gradient** (V1) and one point on it — the **point** P on the curve.

So it is a straight-line question:

$y - y_1 = m(x - x_1)$, then tidy to
 $y = mx + c$.

$$m \approx -0.6, P(2, 2.3)$$

$$y - 2.3 = -0.6(x - 2)$$

$$y = -0.6x + 3.5$$

IT LOOKS LIKE

"Hence find an equation of the tangent"



How do I do it?

1 Gradient

From V1 (your estimate).

2 Point

P — read its y from the curve.

3 Line

$$y - y_1 = m(x - x_1).$$

4 Tidy

$$y = mx + c.$$

GRADIENT + POINT = LINE Your own m carries through.

V2 Equation of a Tangent

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

"Hence find an equation of the tangent to the curve at P" → Use your gradient and P

"Give your answer in the form $y = mx + c$ " → Expand and collect

Your m was an estimate → Follow-through marks: use it consistently

**WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 1HR · Q18(a)(ii) · 2 marks

The gradient of the curve at $P(2, 2.3)$ is about -0.6 . Hence find an equation of the tangent to the curve at P in the form $y = mx + c$.

STEP 1 $m = -0.6, \quad P(2, 2.3)$ ◀ what you know

STEP 2 $y - 2.3 = -0.6(x - 2)$ ◀ point and gradient

STEP 3 $y = -0.6x + 1.2 + 2.3$ ◀ expand

STEP 4 $y = -0.6x + 3.5$ ◀ tidy

ANSWER $y = -0.6x + 3.5$ ✓ $x = 2: -1.2 + 3.5 = 2.3$ ✓**WATCH OUT**

✗ $y - 2.3 = -0.6(x + 2)$

✓ $(x - 2)$

$x_1 = 2$, so the bracket is $x - 2$.

**EXAM TIP**

The line must pass through P : substitute P to check c .

**Check yourself**

calculator first · answers on page 13

2.1

Dr Eslam

The tangent at $P(3, 5)$ also passes through $(0, -1)$. Find its equation.

Answer:

2.2

Practice · Very hard

The gradient of a curve at $(2, 1)$ is 9. Find the equation of the tangent there.

Answer:

2.3

Dr Eslam

The tangent at $(4, 10)$ has gradient -1.5 . Find its equation.

Answer:

V3

VARIANT 3

Zero Gradient: Turning Points

At a maximum or minimum the tangent is flat: gradient 0.

SEEN IN PAST PAPERS

● ● ● ● 1 / 4

PRACTICE QUESTIONS: 2


Your notes

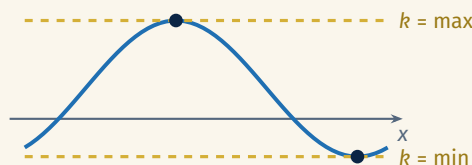
Came up in: Jan 2020 P1HR Q18(b)

? What is it?

In plain words

At the top of a hill or the bottom of a valley the tangent is **horizontal**: gradient **0**.

Between them the gradient is **negative** (going down) or positive (going up). A horizontal line $y = k$ through a turning point **touches** instead of crossing — that changes how many solutions $f(x) = k$ has.



IT LOOKS LIKE

gradient is zero

$$f(x) = k$$

How do I do it?

1 Find

Tops and bottoms of the curve.

2 Read

Their coordinates: gradient 0 there.

3 Sign

Between a max and the next min: gradient negative.

4 $f(x) = k$

Slide a horizontal line; count crossings.

FLAT = ZERO Turning points change the number of solutions.

V3 Zero Gradient: Turning Points

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down the coordinates of the points where the gradient is zero”

→ **The max and min**

“the range of values of x for which the gradient is negative”

→ **Between max and min, going down**

“ $f(x) = k$ has exactly two different solutions”

→ **$y = k$ through a turning point****WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 1HR · Q18(b) · 2 marks

The graph of $y = f(x)$ for $-4 \leq x \leq 12$ has a maximum at $(0, 3)$ and a minimum at $(8, -1)$; $f(-4) \approx -1.2$ and $f(12) \approx 3.8$. The equation $f(x) = k$ has exactly two solutions. Find the two possible values of k .

STEP 1

$y = 3$ touches at $x = 0$, crosses once near $x = 11.7$

◀ at the maximum

STEP 2

⇒ 2 solutions

◀ count

STEP 3

$y = -1$ touches at $x = 8$, crosses once near $x = -3.7$

◀ at the minimum

STEP 4

⇒ 2 solutions

◀ count

ANSWER $k = 3$ and $k = -1$ ✓ other k give 1 or 3 crossings ✓**WATCH OUT**

× counting a touch as two

✓ a touch is one solution

The line meets the curve at one point there.

**EXAM TIP**

Slide a ruler up the page and count crossings — the count changes only at turning points.

**Check yourself**

calculator first · answers on page 13

3.1

Practice · Medium

$y = x^3 - 3x - 2$. Write down the points where the gradient is zero.

Answer:

3.2

Practice · Medium

For $y = x^3 - 3x - 2$, for which x is the gradient negative?

Answer:

3.3

Dr Eslam

Where is the gradient of $y = x^2 - 6x + 5$ zero?

Answer:

V4

VARIANT 4

Rates of Change in Real Life

Gradient of a real graph = rate: units of y per unit of x . Tangent for “at”, chord for “average”.

SEEN IN PAST PAPERS

0 / 4

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 15



Your notes

What is it?

In plain words

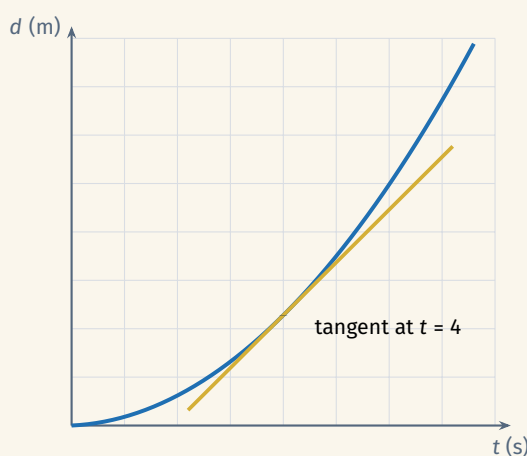
On a **distance-time** graph the gradient is the **speed**; on a **velocity-time** graph it is the **acceleration**.

“At 5 seconds” → **tangent**. “Average over the first 40 seconds” → **chord**. Always say what it means, with units.

IT LOOKS LIKE

speed at $t = 5$

interpret the gradient



How do I do it?

- ## 1 At or average?

“At” → tangent. “Average” → chord.

- ## 2 Draw

The tangent (or chord) on the graph.

- ### 3 Rise over run

With the axis units.

- ## 4 Meaning

Speed (m/s), acceleration (m/s²), rate (°C per hour) ...

SAY IT WITH UNITS “Interpret” wants a sentence.

V4 Rates of Change in Real Life

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Use the graph to estimate the speed at time 5 seconds”



Tangent at $t = 5$ on the distance–time graph

“Interpret the value of the gradient”



“The ... is increasing at ... per ...”

“average acceleration over the first 40 seconds”



Chord from $t = 0$ to $t = 40$

**WORKED EXAMPLE DR ESAM**

Rates of change · 3 marks

On a distance–time graph (metres, seconds), the tangent at $t = 4$ passes through (2, 3) and (6, 17). Estimate the speed at $t = 4$ and say what it means.

STEP 1+2

rise = $17 - 3 = 14$ m, run = $6 - 2 = 4$ s ◀ distance, time

STEP 3

$$\frac{14}{4} = 3.5$$

◀ gradient

STEP 4

at 4 seconds the object is moving at 3.5 m/s

◀ meaning

ANSWER about 3.5 m/s

✓ units: m ÷ s ✓

**WATCH OUT**

× “the gradient is 3.5”

✓ “the speed at 4 s is 3.5 m/s”

Interpret means: what quantity, at what time, with units.

**EXAM TIP**

Read the axis labels first — they tell you the units of the gradient.

**Check yourself**

calculator first · answers on page 13

4.1

Dr Esam

Velocity–time: the tangent at $t = 3$ passes through (0, 10) and (6, 46). Acceleration?

Answer:

4.2

Practice · Very hard

Velocity is 3 m/s at $t = 0$ and 30 m/s at $t = 40$. Find the average acceleration.

Answer:

4.3

Dr Esam

A distance–time tangent at $t = 4$ has gradient 3.5 (m, s). What does it mean?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 about 2.7

1.2 about -0.6

1.3 2

2.1 $y = 2x - 1$

2.2 $y = 9x - 17$

2.3 $y = -1.5x + 16$

3.1 $(-1, 0)$ and $(1, -4)$

3.2 $-1 < x < 1$

3.3 $(3, -4)$

4.1 6 m/s^2

4.2 0.675 m/s^2

4.3 speed 3.5 m/s at 4 s

THE WHOLE TOPIC ON ONE PAGE

Gradients cheat sheet



Your notes
V1 At a point 2/4

$\frac{\text{rise}}{\text{run}}$ of tangent
Long tangent, big triangle.

V2 Tangent line 1/4

$y - y_1 = m(x - x_1)$
Your gradient + the point.

V3 Zero gradient 1/4

max, min
Flat; $f(x) = k$ counts.

V4 Real life

$\frac{m}{s} = \text{speed}$
Tangent "at"; chord "average".

**TOP 3 LOST MARKS**

1. chord instead of tangent ×
2. positive answer on a falling curve ×
3. no units / no meaning ×

BOTTOM LINE Tangent touches once · big triangle · flat = zero · say it with units.